Maamoun MAROUANI

Industrial electronics engineering student | Automation & electrical design (EPLAN)

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Med Maamoun MAROUANI



Summary

Seeking a **4- to 6-month final-year internship starting in February 2027** in electrical design, industrial electronics, or control systems, with a particular interest in electrical schematics, control panels, testing, and commissioning.

I am an industrial electronics engineering student and have already worked on **troubleshooting**, schematic reading, restarting automated equipment, as well as **EPLAN**, **STM32**, **CAN**, and **C++**. I aim to contribute to schematic updates, component selection, and test and validation activities in a hands-on environment.

Professional Experience

LEMAN Industrie Connectique — Tunisia
Engineering internship, maintenance and automation — Jul 1, 2026 - Aug 31, 2026

- Diagnosed faults on a vision inspection machine with conveyor, robot arm, cameras, and an LS PLC, then returned the equipment to service.
- Analyzed mechanical, electrical, and software issues using **schematics**, technical manuals, and machine cycles.
- Updated the PLC logic in **XG5000**, the HMI in **XP-Builder**, and recalibrated the system in Adept.
- Verified operation after intervention and validated repairs to improve equipment reliability.

SYSTELEM INDUSTRIE
Introductory internship — Jul 15, 2025 - Aug 15, 2025

- Inspected machined parts and observed the operation of Biglia and Citizen Cincom CNC machines.
- Participated in electromechanical maintenance and an introduction to **G-code** to understand machine cycles.

Education

École Nationale d'Ingénieurs de Sousse (ENISo) — National engineering degree in industrial electronics engineering
Industrial Electronics Engineering — 2024 - Present

Institut Préparatoire aux Études d'Ingénieurs de Kairouan (IPEIK) — Preparatory cycle – DEUPC
Mathematics and Physics — 2022 - 2024

Projects

BMS on STM32
Designed a measurement and protection system for a Li-ion battery on STM32, with passive balancing, state-of-charge estimation, and UART/I²C communication. May - Jun. 2026.

Battery cell monitoring and CAN communication
Acquired voltage and temperature data, detected anomalies, and validated CAN transmission in a battery cell monitoring system. Jul. - Sept. 2026.

Industrial electrical monitoring (STM32)
Built a prototype for measuring electrical quantities with STM32, Modbus RTU, I²C, display, and alerts for electrical monitoring. Mar. - Jul. 2026.

Automotive diagnostic system – OBD-II
Read and visualized vehicle data in real time via an ELM327 module, with diagnostic logic and embedded data acquisition. Dec. 2025 - May 2026.

Smart IoT monitoring system – ESP32
Detected gas and noise with an ESP32 system, web interface, and sub-second latency for real-time monitoring. Feb. - Mar. 2025.

Autonomous obstacle-avoiding robot
Developed an autonomous robot that avoids obstacles using an HC-SR04 ultrasonic sensor. Oct. - Dec. 2024.

Languages

French
Fluent

English
Fluent

Spanish
Basic knowledge

Skills

Electronics and design
EPLAN, FPGA, VHDL, Altium Designer, KiCad, EPLAN, Proteus

Automation and control systems
PLC, Ladder, HMI, GRAFCET, Industrial instrumentation, XG5000, XP-Builder

Programming and tools
C, C++, Python, G-code, MATLAB/Simulink

Industrial communication and data acquisition
CAN, CAN-FD, UART, Modbus RTU, Modbus ASCII, FlexRay, Bluetooth, Wi-Fi

Embedded systems and programming
STM32, HAL, ESP32, Arduino, DSP, STM32CubeIDE, Arduino IDE

Additional Training

DRÄXLMAIER × ENISo
Ongoing technical training — Automotive wiring harnesses, low- and high-voltage architectures, and onboard diagnostics — Oct. 2026 - Present

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